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EXPERIMENT 13

Perspective Transformation on a Video

AIM

To perform a perspective transformation on each frame of a video (bird's-eye view).

PROGRAM

```
import cv2
import numpy as np

cap = cv2.VideoCapture("sample_video.mp4") # use 0 for the web camera

# Region of interest in the frame -> rectangle in the output
roi_points = np.float32([(150, 120), (490, 120), (590, 440), (50, 440)])
target_points = np.float32([(0, 0), (400, 0), (400, 600), (0, 600)])
M = cv2.getPerspectiveTransform(roi_points, target_points)

while True:
    ret, frame = cap.read()
    if not ret:
        break

    dst = cv2.warpPerspective(frame, M, (400, 600))

    cv2.polylines(frame, [roi_points.astype(np.int32)], True, (0, 255, 0), 2)
    cv2.imshow("Original Frame", frame)
    cv2.imshow("Transformed Frame", dst)

    if cv2.waitKey(25) & 0xFF == ord("q"):
        break

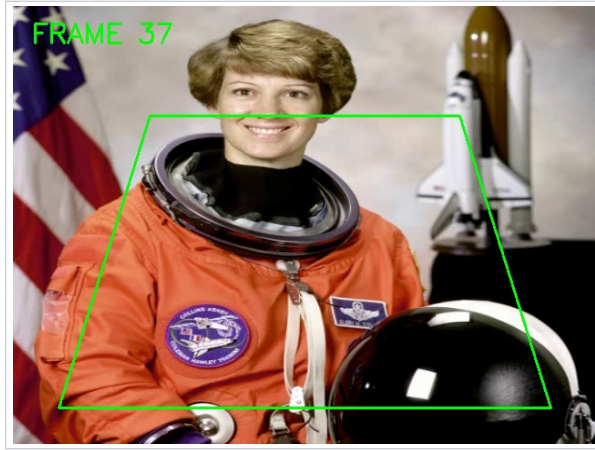
cap.release()
cv2.destroyAllWindows()
```

INPUT



sample_video.mp4 (input video, 640 x 480)

OUTPUT



Original frame with the region of interest marked in green



The same frame after `cv2.warpPerspective()`: the marked region is warped to a 400 x 600 rectangle

RESULT

Thus a perspective transformation was successfully applied to every frame of the video.